

Progress report 05

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Work log

Date	Number of Hours	Description of work done
Sep 07, 2025	1.5	Reviewed the document General Guidelines and Deliverables provided by the instructor
Sep 08, 2025	3	Reading the A Guide to Technical Report Writing to be familiar with the rules
Sep 08, 2025	2	Drafted the proposal for my part of the project, which focuses on the development of the website portal for displaying and analyzing game data
Sep 09, 2025	3	Worked on project planning and created the Gantt chart for the website portal tasks using Tableau. Updated the proposal with the timeline section.
Sep 11, 2025	6	Compared several data visualization libraries such as Highcharts, Plotly, D3.js and Chart.js. I tested basic charts in each library to see how they work with PHP/MySQL. Highcharts and Plotly had many advanced features but needed paid licenses or extra setup for some functions. D3.js was powerful but very hard to learn and required a lot of custom coding. Chart.js was easier to use, had clear documentation, and worked smoothly with PHP and MySQL. Because of this I finally chose Chart.js for our project.
Sep 12, 2025	2.5	Revised the project proposal based on instructor's feedback, including unifying the narrative perspective, adding detailed deliverables, and correcting tense and grammar issues in the report.
Sep 13, 2025	3	Discussed data structure issues with teammates, finalized basic tables and related columns, and aligned on game mechanics and design approach.
Sep 15, 2025	4	Researched server options for handling game data uploads; compared multiple online server

		providers but have not finalized the choice yet.
Sep 18, 2025	4	Set up the basic environment with PHP and MySQL database; used local resources temporarily since the server has not been finalized.
Sep 20, 2025	1	Revised the project proposal again, reorganized the Deliverables section for clarity and detail.
Sep 22, 2025	3	Discussed data structure with teammates, confirmed the columns of the match table, and analyzed how the portal should retrieve data and the basic API design.
Sep 25, 2025	2	Revised GitHub repository settings based on instructor's requirements; renamed repository and updated README.md file accordingly.
Report2		
Sep 27, 2025	3.5	Finished choosing the server and decided to use Google Cloud. Created a new VM and set up PHP, Apache, and MySQL on it.
Sep 29, 2025	4	Installed the needed PHP extensions. Tested the connection between Apache and MySQL, and made sure PHP pages can get and show data from the database.
Oct 1, 2025	3.5	Built the static HTML/CSS structure of the portal as a test and prototype, including homepage layout, navigation bar, and chart area. Uploaded the static version to GitHub
Oct 2, 2025	2	Discussed game data, table design, and website functions with teammates. Some planned functions are not supported by the game yet, so they are left empty for now.
Oct 3, 2025	4.5	implemented PHP backend functions to connect to MySQL. Designed page logic and tested data reading and display successfully.
Oct 6, 2025	4	Developed RESTful API for CRUD operations. Tested the API with Postman locally and on the cloud server. Uploaded the world-balance-api project to GitHub.
Oct 8, 2025	3.5	Deployed the full portal website on the cloud server. Tested the connection between front-end and back-end. Started adding sample data into the database for testing (still in progress).
Report3		
Oct 10, 2025	3.5	Started research on Chart.js basic usage.

		Learned how to draw bar and line charts using sample data. Practice a simple HTML + JS page and upload the github
Oct 12, 2025	4	Integrated Chart.js with PHP + MySQL. Built API to send json data to the chart. Tested fetching data from the database and showing
Oct 13, 2025	3	Wrote the Applied Research Report: Using Chart.js for Data Visualization in a PHP + MySQL Web Portal.
Oct 15, 2025	4.5	Continued the second research about Chart.js interaction. Learned to use events like onClick and onHover. Tested changing datasets by button click and chart.update(). (not complete)
Oct 17, 2025	4	Applied the interaction functions to the Players page. When selecting different players, all charts update automatically. Adjusted chart colors
Oct 19, 2025	3.5	Finished the main layout and data display for the Vehicles page. Added filters for nation and tank type, and a search box for vehicle name.
Oct 21, 2025	2	Improved both Players and Vehicles pages. Added third chart "Top 10 Players by Win Rate" with horizontal bars. Verified all charts refresh when player changes
Report4		
Oct 23, 2025	2.5	Debugged minor chart refresh issues when switching between players.
Oct24,2025	5	Updated PHP backend queries to include additional player statistics; Prepare for midterm report
Oct29,2025	4	Adjusted chart container responsiveness for different screen sizes.
Nov3,2025	2	Met with teammates to confirm data synchronization between the game server and portal database. Discussed possible structure of final presentation.
Report5		
Nov 10,2025	3	Finished setting up the MySQL database on the cloud server. Tested the connection, fixed a few connection errors and confirmed that all data can be read and written correctly.
Nov 12,2025	3.5	Added session management in PHP to keep

		user login status. Tested login and logout functions with Firebase authentication.
Nov 13,2025	3	Implemented the access control feature. Now only logged-in users can see detailed data on the Vehicles page, add sort functionality
Nov 14,2025	2.5	Updated the Home page search function
Nov 17,2025	5	Added a new Compare function on the Players page which is based whether users login or not
Nov 19,2025	2	Met with teammates to discuss the final database access structure. We talked about how the portal should share and sync data. Also planned the final presentation

Description of work done

During this period, I mainly focused on completing the new features and improving the overall user experience of the website portal.

I finished setting up the cloud database and tested the connection between the portal and the server to make sure all data loads correctly.

I also added session handling and access control and firebase authentication, so only login users can see extra information on the Vehicles page, such as predicted win rate and efficiency score.

In addition, I updated the search function on the Home page.

Now users can type either a player name or a vehicle name, and the system will automatically go to the correct page.

Another important part was adding the new Compare function on the Players page. Login users can now compare two players on one chart and choose what data they want to compare. It makes the portal more interactive and useful for analysis.

Finally, I met with my teammates to discuss the final structure of database access and how our pages should interact with the backend.

We also talked about our plan for the final presentation and divided the work for the last stage of the project.

Repo Check-in of Implementation completed

In this phase, I added and fixed several files to the Implementation section in GitHub.

The files/folders I have checked in the repo are as follows:

- Implementation/portal/portalDemo/assets/css (fixed all problem on styles)
- Implementation/portal/portalDemo/players.php (add compare functionality and access control)
- Implementation/portal/portalDemo/includes/header.php(added access control and

firebase authentication on header in each page)

- Implementation/portal/portalDemo/includes/LoginManager.php
- Implementation/portal/portalDemo/includes/db.php (fixed the database connection to cloud)